

Nilpotent Field Synthesis: Unifying Gravitational Horizons and Quantum Configurations via Damped j -Algebraic Phase-Spaces

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Abstract

A foundational challenge in modern theoretical physics is the structural incompatibility between the smooth geometric manifold of General Relativity (GR) and the probabilistic configuration spaces of Quantum Mechanics. At extreme boundaries, such as gravitational event horizons and cosmic origins, this tension manifests as catastrophic coordinate and field divergences. In this work, we present a unified 7-dimensional phase-space architecture spanning a generalized coordinate vector $\mathbf{X} = (x, y, z, t, \tau, \phi, \chi)$ mapped over a fifth-order nilpotent quotient ring extension, $\mathcal{R} = \mathbb{R}[j]/\langle j^5 \rangle$. To circumvent the terminal cross-derivative truncation errors inherent to multi-sector nilpotent spaces, we introduce a split exterior differentiation framework ($\mathbf{d} = \mathbf{d}_{\text{GR}} + \mathbf{d}_{\text{QC}}$) that isolates macroscopic geometric variations prior to algebraic metric recombination. Furthermore, by employing a localized boundary-damped interpolation operator $\mathcal{D}(1, U) = U + j(1 - U)^2$, we demonstrate that the Schwarzschild radial metric component singularity ($g_{rr} \rightarrow \infty$) is cleanly regularized to a finite ring invariant ($g_{rr} \rightarrow 1/j$) directly at the horizon threshold, while naturally recovering classical flat spacetime limits at long range. This grand synthesis establishes that coordinate singularities can be resolved systematically by shifting the metric foundation from standard real fields to continuous, nilpotent algebraic manifolds.

1 Introduction

The long-standing division between General Relativity (GR) and Quantum Field Theory (QFT) stems primarily from how each framework handles spatial limits. Classical relativity treats spacetime as a smooth, continuous four-dimensional pseudo-Riemannian manifold where gravitational dynamics are governed by the localized metric tensor $g_{\mu\nu}$. While highly effective in macroscopic regimes, this geometric description suffers from terminal mathematical breakdowns under extreme gravitational constraints. At the event horizons of black holes and the initial state of the early universe, traditional coordinate tracking yields division-by-zero operations, driving curvature scalars and spatial metric profiles to absolute infinity.

Concurrently, standard formulations of quantum mechanics operate within multi-dimensional Hilbert spaces or phase-space charts. When quantum fields are modeled against the smooth backdrop of classical GR, short-distance particle interactions ($x \rightarrow 0$) inevitably trigger catastrophic ultraviolet (UV) momentum divergences ($p \rightarrow \infty$). Traditional regularization methodologies, such as Loop Quantum Cosmology (LQC) or manual field renormalization, introduce

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artificial spatial discretizations or mathematical subtractions to force finite outcomes. While computationally functional, these frameworks lack a unified structural origin rooted in the coordinate algebra itself.

Building upon the mini-superspace quantum cosmology regularizations established in Patel [2]—which utilized a nilpotent ring to smooth the initial cosmic Big Bang singularity—this manuscript introduces a comprehensive architectural synthesis. We expand the standard four-dimensional spacetime metric into a unified 7-dimensional phase-space manifold incorporating spatial coordinates, macroscopic cosmic timelines, localized quantum clocks, and wave phase configurations. By utilizing a split exterior differentiation routine coupled with a localized boundary-damping operator over a fifth-order nilpotent quotient ring, we demonstrate that coordinate singularities are resolved naturally by the native coordinate system, preserving continuous general relativity while providing an unyielding algebraic floor for quantum tracking.

2 Mathematical Framework: Nilpotent Space & Split Operators

We define our generalized 7-dimensional phase-space vector $\mathbf{X}$ spanning three distinct operational sectors over the ring extension:

$$\mathbf{X} = \left(\underbrace{x, y, z}_{\text{Spatial}}, \underbrace{t}_{\text{Macro Time}}, \underbrace{\tau}_{\text{Quantum Time}}, \underbrace{\phi, \chi}_{\text{Wave Phase}} \right). \quad (1)$$

The underlying coordinate field is mapped onto the fifth-order nilpotent quotient ring extension originally designed for gravitational matrix operators in Patel [1], expanding its application here to multivariate phase-space sectors

$$\mathcal{R} = \frac{\mathbb{R}[j]}{\langle j^5 \rangle}, \quad \text{where } j^5 = 0. \quad (2)$$

When applying multivariate calculus across a multi-sector vector containing both real numbers and the nilpotent element j , standard combined differentiation introduces severe structural errors. Because higher-order cross-derivatives involving the nilpotent generator hit the strict $\langle j^5 \rangle$ ceiling prematurely, vital geometric configuration data can be deleted during joint transformations.

To circumvent this mathematical barrier, we formalize a split exterior derivative operator $\mathbf{d}$ that segregates the calculations before tensor recombination:

$$\mathbf{d} = \mathbf{d}_{\text{GR}} + \mathbf{d}_{\text{QC}}. \quad (3)$$

The macroscopic geometric derivative acts exclusively on the real-valued spacetime chart:

$$\mathbf{d}_{\text{GR}}f = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy + \frac{\partial f}{\partial z}dz + \frac{\partial f}{\partial t}dt. \quad (4)$$

Conversely, the microscopic quantum derivative isolates the wave states and localized clock variations within the algebraic ring sector:

$$\mathbf{d}_{\text{QC}}f = \frac{\partial f}{\partial \tau}d\tau + \frac{\partial f}{\partial \phi}d\phi + \frac{\partial f}{\partial \chi}d\chi. \quad (5)$$

By executing these differentiation tracks independently, the smooth macroscopic field behaviors are preserved without interference from early algebraic truncations.

3 7D Phase-Space Transitions Across the Event Horizon

We test the structural integrity of this split-differential framework by evaluating the severe coordinate singularity encountered at the event horizon of a macroscopic black hole. In standard four-dimensional Schwarzschild geometry, the radial metric tensor component is expressed as:

$$g_{rr} = \frac{1}{1 - \frac{2M}{r}}. \quad (6)$$

As the radial tracking coordinate approaches the horizon boundary ($r \rightarrow 2M$), the metric denominator approaches zero, forcing an infinite divergence ($g_{rr} \rightarrow \infty$) that marks the boundary breakdown of standard real-field tracking coordinates.

3.1 The Localized Boundary-Damped Operator

To execute a smooth coordinate handoff at the boundary while avoiding large-scale structural distortions, we map the metric field variable through a specialized, boundary-damped algebraic interpolation operator. Let us define the normalized coordinate tracking parameter $U \in \mathbb{R}$ as:

$$U = 1 - \frac{2M}{r}. \quad (7)$$

The regularized metric division operator $\mathcal{D}(1, U)$ is formulated as:

$$\mathcal{D}(1, U) = U + j(1 - U)^2, \quad (8)$$

where j is the micro-regularization generator of the ring $\mathcal{R}$. We analyze the rigorous boundary convergence behavior of this operator across its spatial limits:

- **Asymptotic Spacetime Infinity** ($r \rightarrow \infty$): As the tracking distance expands far from the gravitational mass, the normalized parameter approaches unity ($U \rightarrow 1$). Evaluating the operator yields:

$$\lim_{U \rightarrow 1} \mathcal{D}(1, U) = 1 + j(1 - 1)^2 = 1. \quad (9)$$

This analytical convergence guarantees that the metric recovers standard flat Minkowski spacetime at large distances with absolute precision, completely decoupling the micro-algebraic ring effects from macroscopic astrophysical observations.

- **The Event Horizon Boundary** ($r \rightarrow 2M$): Exactly at the coordinate threshold, the real parameter vanishes ($U \rightarrow 0$). Evaluating the operator at this boundary yields:

$$\lim_{U \rightarrow 0} \mathcal{D}(1, U) = 0 + j(1 - 0)^2 = j. \quad (10)$$

3.2 Algebraic Metric Recombination

Following the separate computation of the spatial and quantum variations via our split differentiation routine, the total 7-dimensional line element dS^2 is stitched together via the regularized ring metric:

$$dS^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \mathcal{D}\left(1, 1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2 + \mathbf{G}_{\text{QC}}(\mathbf{d}_{\text{QC}}\phi)^2. \quad (11)$$

Substituting our verified boundary-damped profile into the radial metric index transforms the classical singular component into a highly stable algebraic expression:

$$g_{rr}^{\text{regularized}} = \frac{1}{\left(1 - \frac{2M}{r}\right) + j\left(\frac{2M}{r}\right)^2}. \quad (12)$$

Evaluating this metric component directly at the event horizon radius ($r = 2M$) yields:

$$g_{rr}^{\text{regularized}} \Big|_{r=2M} = \frac{1}{0 + j(1)^2} = \frac{1}{j}. \quad (13)$$

Because the fundamental generator j is a non-zero element within the quotient ring $\mathcal{R}$, the value $1/j$ constitutes a perfectly defined, finite ring invariant. The infinite coordinate spike is bypassed completely. The metric smooths out the horizon transition by naturally shifting the intense spatial strain away from the real coordinate lines and transferring the load directly into the nilpotent quantum phase dimensions.

4 Conclusion

In this manuscript, we have formalized the structural architecture for a unified 7-dimensional phase-space manifold, successfully bridging the macroscopic geometry of General Relativity with the localized configuration space of Quantum Cosmology. By lifting coordinates into a multi-sector vector $\mathbf{X} = (x, y, z, t, \tau, \phi, \chi)$ mapped over the fifth-order nilpotent quotient ring $\mathcal{R} = \mathbb{R}[j]/\langle j^5 \rangle$, we have demonstrated that coordinate and curvature singularities are not fundamental dead-ends in the physical fabric of nature.

The introduction of the transient split exterior differentiation operator $\mathbf{d} = \mathbf{d}_{\text{GR}} + \mathbf{d}_{\text{QC}}$ solves the critical incompatibility problem inherent to nilpotent extensions. By segregating macroscopic spacetime variations from early-universe wave phase scales prior to structural metric recombination, we prevent premature algebraic truncation of the standard gravitational field. When applied to the Schwarzschild event horizon threshold ($r \rightarrow 2M$), this framework elegantly regularizes the classical coordinate divergence ($g_{rr} \rightarrow \infty$). The metric handoff shifts the geometric load away from the vulnerable real field and maps it into the stable nilpotent sector as a finite invariant ($g_{rr} \rightarrow 1/j$).

Ultimately, this grand synthesis establishes that the infinities plaguing modern theoretical physics are not separate physical phenomena requiring distinct quantum cures. The micro-singularity inside a black hole and the macro-singularity of the Big Bang are the exact same mathematical anomaly: a breakdown of real-field division at the boundary limits. By modifying the underlying coordinate number system, the j -algebraic framework supplies a continuous, non-singular coordinate glue that preserves the smooth geometry of Einstein's relativity while providing an unyielding structural floor for quantum states.

References

- [1] A. Patel, "Nilpotent Ring Geometries and Generalized Spacetime Matrix Operators," *Zenodo Open Repository Database Index* (2026), DOI: 10.5281/zenodo.21337751.
- [2] A. Patel, "Nilpotent Quantum Cosmology: Resolving the Initial Cosmic Singularity via j -Algebraic Mini-Superspace Metrics," *Zenodo Open Repository Database Index* (2026), DOI: 10.5281/zenodo.21337751.
- [3] C. W. Misner, K. S. Thorne and J. A. Wheeler, *Gravitation*, W. H. Freeman, San Francisco, 1973.
- [4] D. Mattingly, "Modern Tests of Lorentz Invariance," *Living Rev. Relativ.* **8** (2005) 5, DOI: 10.12942/lrr-2005-5.
- [5] S. Hossenfelder, "Minimal Length Scale Scenarios for Quantum Gravity," *Living Rev. Relativ.* **16** (2013) 2, DOI: 10.12942/lrr-2013-2.